电子科技大学
格拉斯哥学院
Glasgow College, UESTC

Logbook

From:2025/10/01 To:2025/10/31

Month	**List the main activities** (only few words per activity)	**Interaction with the supervisor**			**Any other form of supervisory interaction** (second supervisor, industry, fellows etc.)
		Number of meetings	Mode of meeting (face- to-face, online e.g., Skype, WeChat etc.)	Number of emails exchanged	
October	1. PyTorch Fundamentals (10/1-10/7): - Read PyTorch technical documentation. - Learned data loading, environment configuration, and CNN concepts (convolutional/pooling layers). - Practiced processing the CIFAR dataset. 2. U-Net Deep Dive (10/9-10/17): - Systematically studied the U-Net paper. - Learned U-Net environment setup and	1	face- to-face	3	Second supervisor: Liu ZhiShuai

	dataset processing through tutorials. - Prepared and delivered a presentation for the group meeting on 10/17. 3. **Framework Focus (10/17-10/24):** - Researched U-Net variants (U-mamba, U-Net++, ResUNet). - Identified nnU-Net as an adaptive framework and decided to make it the research focus. - Downloaded nnU-Net v2 source code and set up a new environment. - Resolved configuration issues (Nvidia driver version, v1/v2 naming differences, 3D Nii data format). 4. **Baseline Model Training & Debugging (10/25-10/26):** - Followed supervisor's guidance to split the dataset (train/validation/test). - Set max epochs to 150 and started training. - Discovered that test set metrics (mIoU,				

	Recall) were extremely low. - After a reminder from the supervisor, investigated and confirmed the issue was caused by image flipping during nnU-Net v2 data augmentation. - Obtained baseline evaluation results (mIoU: 0.8107). 5. **Experimental Iterations (10/27-10/28):** - **(10/27)** Discussed results with supervisor; retrained using the CLAHE-enhanced dataset. Results showed no significant improvement. - **(10/28)** Reported results; per supervisor's guidance, changed the loss function to Focal Loss and retrained. Results still showed no significant change.				
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